

EXPERIMENT 8 — DOCUMENT CONTEXT

Ex. No.: 8

Title: Use StegExpose to Detect Hidden Data in Images

Date: _____

Aim

To use **StegExpose** to analyze digital images and detect possible hidden data using statistical steganalysis.

Objective

- To examine digital images for possible steganographic content.
- To analyze the statistical properties of the given images using StegExpose.
- To obtain the **Fusion score** for each image.
- To compare the obtained scores with the default steganographic detection threshold.

Software / Tools Required

- Java Runtime Environment (JRE)
- StegExpose
- Windows Command Prompt
- Test images

Procedure

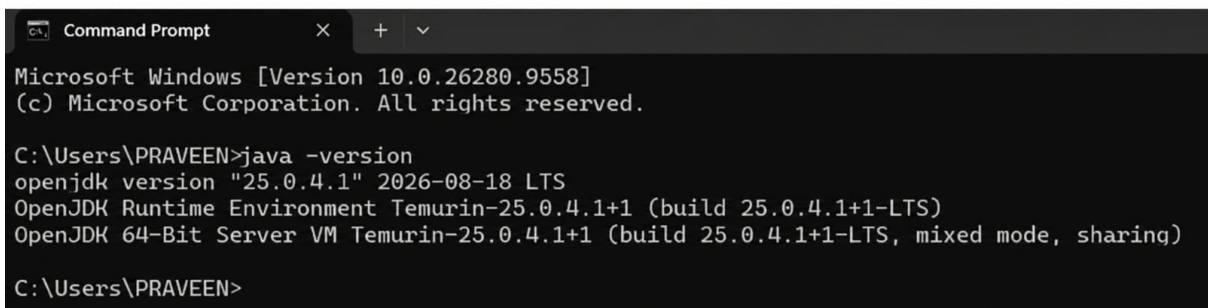
Step 1: Verify Java Installation

Java installation was verified using the following command:

```
java -version
```

The installed Java environment was displayed successfully, confirming that Java was available for executing the StegExpose JAR file.

Screenshot 1 — Java version verification



```
Microsoft Windows [Version 10.0.26280.9558]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PRAVEEN>java -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)

C:\Users\PRAVEEN>
```

Step 2: Create the Experiment Directory

A separate directory was created on the D: drive for Experiment 8.

D:

```
mkdir DF_Experiment_8
```

```
cd DF_Experiment_8
```

The experiment directory used was:

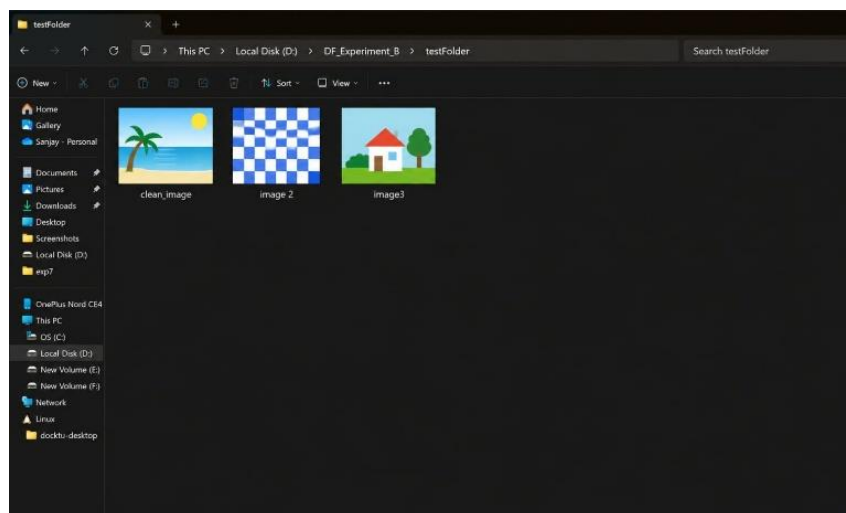
```
D:\DF_Experiment_8
```

A subdirectory named testFolder was created to store the test images.

```
mkdir testFolder
```

The StegExpose JAR file and test images were placed in the experiment directory and testFolder, respectively.

Screenshot 2 — Experiment folder and test images



Step 3: Execute StegExpose

The StegExpose program was executed on the test image folder using:

```
java -jar StegExpose.jar testFolder
```

The command completed successfully without displaying an error.

Screenshot 3 — StegExpose execution

```
D:\DF_Experiment_8>D:
D:\DF_Experiment_8>cd \DF_Experiment_8
D:\DF_Experiment_8>java -jar StegExpose.jar testFolder
D:\DF_Experiment_8>java -jar StegExpose.jar testFolder default default results.csv
```

Step 4: Generate the Steganalysis Results

To obtain detailed statistical results in CSV format, the following command was executed:

```
java -jar StegExpose.jar testFolder default default results.csv
```

The command generated the file:

results.csv

The results were then displayed using:

Step4 : Analyze the Results

The generated CSV file contained the following results:

```
D:\DF_Experiment_8>type results.csv
File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
clean_image.png,false,2,0,00135560875678536,0,0,0.0010629026672680287,0.0014619573339978242,0.701162192340973E-4
image_2.png,false,0,0,0,0,0,0,0,0,0,0
images_3.png,false,10,0,00750732396181886,0,0,0.009029365936654453,0.008589190195225856,0.006281470023424792

D:\DF_Experiment_8>
```

The default StegExpose threshold used in the experiment was **0.2**.

All three images produced Fusion scores below **0.2**, and the Above stego threshold? field was reported as **false** for all three images.

Screenshot 4— results.csv output and Fusion scores

[illegible]

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

StegExpose was successfully used to perform statistical steganalysis on the three test images. The generated results.csv file provided the statistical analysis values and Fusion scores.

All three images were reported as **False** under Above stego threshold?, indicating that none of the analyzed images exceeded the configured StegExpose threshold of **0.2**.